



Student Assessment Information Sheet

This document is for University of Dundee students for their use in completing assessed work. It should not be passed to third parties or posted on any other website.

Assessment Information	
Module Name:	DevOps and MicroServices
Module Code:	AC51041/GA42010
Assessment Title:	System Design Brief
Assessment Weighting:	20%
Assessment Due Date	Sunday 16 th November 2025

If you have questions about this assessment, please contact one of the staff on the module

Learning Outcomes
This assessment addresses the following learning outcomes of the module: - Compare and contrast technologies used to develop microservices systems: You will demonstrate this by evaluating different technologies in your architectural design, justifying your choices for each microservice component. - Understand the principles of continual development: You will show this through a well-structured continual development plan (e.g., CI/CD pipeline), ensuring the system is able to evolve smoothly with minimal downtime. - Develop a system using NoSQL: You will include NoSQL databases in your design, providing a comparison of NoSQL and relational databases, and explaining why NoSQL might be suitable for this microservices architecture. - Implement a variety of tools to develop a cloud-based microservices system: You will identify and justify tools such as containerisation, orchestration, and cloud services in your design, showing how they interact to create a scalable, cloud-based system. - Develop and apply problem-solving, team working, communication, time-management, self-assessment, and independent study skills: You will demonstrate this by producing a clear, well-organised report that effectively communicates your design and its justifications.

AI Usage Guidance
Given the broad and evolving nature of these tools, it is important to be clear about what is permitted in each assessment. Your module lecturers will indicate to what extent generative AI can be used. If

you are unsure whether the use of a specific AI tool is permitted, please contact your module coordinator for guidance. Misuse or failure to disclose AI assistance may be treated as academic misconduct and subject to procedures outlined in the [academic misconduct by students code of practice](#).

The levels of AI usage in assessment are:

RAG Rating	Permitted Level	Description
Red (Forbidden)	No	AI tools must not be used for this assessment. This includes generating content, conducting analysis, or providing answers to specific questions.
Amber (Restricted)	No	AI tools may be used with caution, but their use must be explicitly acknowledged. For example, AI assistance in brainstorming, drafting, or structuring ideas is allowed, but the final submission must be your own work. Any AI-generated content must be fully reviewed, edited, and critically evaluated.
Green (Permitted)	Yes	AI tools are permitted for this assessment without restriction, but their use must be explicitly acknowledged. You may use AI systems for research assistance, content generation, and other purposes, provided that the usage aligns with the learning outcomes and your understanding is demonstrated. You should engage critically with output and apply your own judgement.

Assessment Task

You have been tasked to design a scalable conference room booking system by a company that provides **conference rooms** services in various **locations** across the country. Each location may have one or more conference rooms. Every conference room must specify its capacity (maximum number of guests it can hold), and its base price. Conference rooms can be booked once per day (that is, for a whole day).

Conference room temperatures are usually held at 21 degrees C to keep the guests comfortable, so when the **user** is booking a conference room, the forecasted weather for that day affects the final price of that booking to account for the additional cost of heating or cooling.

Forecasted temp. difference (from 21 deg C)	Additional charge (on top of the base price)
>= 0 but less than 2 degrees C	No additional charge
>= 2 but less than 5 degrees C	10%
>= 5 but less than 10 degrees C	20%
>= 10 but less than 20 degrees C	30%
>= 20 degrees C	50%

You are required to create the weather forecast system as a separate web service which takes a location id or name, and date, and returns the forecasted weather for the given day (in JSON format). You should simulate the forecasted weather data. There is no requirement to use a real weather service here.

Please note that users of this system must be authenticated in order to book a conference room. You should pre-populate the system with reasonable/relevant data including conference room data, location data, etc.

You can make assumptions about the system where there is missing information but please discuss these in your design report.

Your role for this component is to produce a **design report** detailing the architectural choices, tools, and technologies you would use to build this system. Your report should have the following structure:

1. Architectural Design (50% of Design Report):

- Describe the high-level architecture of the system, using microservices principles.
- Identify key components such as:
 - User Authentication Service
 - Weather Forecast Service
 - Etc.
- Include diagrams that illustrate the data model (e.g., UML diagrams or other as applicable), and how these services communicate (using cloud services like AWS where appropriate).
- Justify the choice of technologies for the services (e.g., NoSQL for data storage, REST APIs for service communication).

2. Technology Stack (30% of Design Report):

- Compare and contrast at least **two different technologies** for each major system component (e.g., MongoDB vs PostgreSQL for user data storage, Express.js vs Flask for web services).
- Highlight the trade-offs in using each technology, focusing on scalability, ease of use, and integration with cloud services like AWS.
- Explain the choice of cloud services (e.g., using EC2 for hosting microservices, etc.).

3. Continual Development Plan (20% of Design Report):

- Outline a high-level **CI/CD** (Continuous Integration and Deployment) pipeline to

- ensure the system can be deployed and updated regularly without downtime.
- Discuss the tools you would use for automation (e.g., Jenkins, GitLab CI/CD).
 - Explain the testing strategy to ensure each microservice is thoroughly tested before deployment.

Deliverables

A written design report (maximum 4 pages) including:

- **Introduction** explaining the system objectives.
- **Architectural Diagrams** showing the relationships between the system components.
- **Technology Comparison Table** for each service, outlining the pros and cons.
- **Justifications** for the chosen technologies and services.
- **CI/CD Pipeline** plan with tools, processes, and rationale.

The report must be submitted as a word or PDF document.

How to Submit

Assessments should be submitted to My Dundee in the 'Assessment and Feedback' area. Please only submit a single word or PDF document.

Assessment Feedback

Feedback on your assessment will be provided via **My Dundee** and will be available within **three weeks** of the submission deadline. Any alteration to this will be communicated with you in advance through My Dundee.

Feedback can be provided in multiple ways. This can include informally during lab and tutorial sessions (e.g. discussing work and sample solutions being explored), and formally after marking (e.g. written, oral, video-based).

Academic Integrity

By submitting this Assessment, you are bound by the University of Dundee policies in relation to Academic Standards and Integrity. Please refer to [Policy Statements, Regulations and Guidance](#) for more information

Late Penalties

One grade point per day late (meaning if a submission is one day late and marked as a C2 it will receive a C3 grade). A day is defined as each 24-hour period following the submission deadline including weekends and holidays. Assessments submitted more than 5 days after the agreed deadline will receive a zero mark (AB).

Generic Assessment Criteria		
A1 – A5	Meets learning outcomes	Excellent work demonstrating a comprehensive understanding, creativity, and advanced critical/analytical abilities. Substantial research well beyond the recommended minimum, using diverse methodologies. Exceptional grasp of knowledge and discipline-specific theories. Shows creative thinking, originality, and independent work. Exemplary use of learning resources. Exhibits highly developed problem-solving skills. Work produced with exceptional accuracy, proficiency, and independence. Superior communication and expression.
B1 – B3		Very good work, showing a solid understanding of theories, concepts, and issues, with clear critical analysis. Research goes beyond the recommended minimum, using established methodologies accurately, with minimal irrelevant content. Very good understanding of knowledge and discipline-specific theories, with some originality and independence. Very good use of learning resources. Displays well-structured arguments and interpretation. Demonstrates creativity, originality, and problem-solving abilities. Work produced with high accuracy, proficiency, and independence. Effective communication and expression, with a solid professional skill set.
C1 – C3		Good work showing an understanding of relevant theories, concepts, and issues, with some critical analysis. Research is carried out accurately using established methodologies, occasionally extending beyond the minimum required. Some errors or irrelevant content may be present. A good understanding of knowledge and discipline-specific theories, with some signs of originality and independence. Good use of learning resources. Provides logical arguments and interpretation with supporting evidence. Shows some creativity and problem-solving skills, though inconsistencies may appear. Expression and presentation are generally accurate, proficient, and carried out with some independence. Good communication and expression with an appropriate professional skill set.
D1 – D3		Adequate work showing a basic understanding of relevant theories, concepts, and issues. Demonstrates sufficient understanding of knowledge and discipline-specific theories to address key concepts. Some errors or gaps in research may be present, and irrelevant content is likely. Basic use of learning resources. Shows some ability to formulate and sustain an argument. Displays some creativity and problem-solving abilities, though inconsistencies are present. Expression and presentation meet minimum standards for accuracy. Communication and expression are sufficient.
MF1 – MF3	Fails to meet learning outcomes	Does not meet the required outcomes. Shows a limited understanding of relevant theories, concepts, and issues. Minimal evidence of research or use of established methodologies. Some relevant content may be included, but there are significant gaps in analysis. Fundamental errors and misunderstandings are likely. Limited use of learning resources. Arguments are weak and poorly developed. Little originality, creativity, or problem-solving ability is evident. Expression and presentation fall short of accuracy and proficiency. Communication and expression are insufficient, with gaps in professional skills.

BF - CF		Does not meet outcomes. Demonstrates a lack of understanding of relevant theories, concepts, and issues. Minimal research and use of established methodologies, leading to incomplete knowledge of the subject. Significant errors and missing components are apparent. Little to no use of learning resources. Arguments are very weak, with no indication of alternative perspectives. Little to no evidence of originality, creativity, or problem-solving skills. Expression and presentation do not meet the necessary standards for accuracy and proficiency. Communication and expression are severely lacking, with significant gaps in professional skills.
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